

(6.3(a) continued)

$$4. \frac{\frac{2}{x+2}}{\frac{1}{x+2} + \frac{2}{x}}$$

LCD = _____

$$5. \frac{4 - \frac{1}{x^2}}{4 + \frac{4}{x} + \frac{1}{x^2}}$$

LCD = _____

$$6. \frac{\frac{x+4}{5} - 3}{\frac{1}{2} + \frac{x+4}{10}}$$

LCD = _____

6.3(b) Using Complex Fractions

Follow the steps to simplify the expression below, then **refer to this worksheet while doing tonight's homework**.

$$\frac{\frac{x-4}{x^2-9}}{1+\frac{1}{x-3}} - \frac{1}{x-3}$$

Step 1: Simplify the complex fraction $\frac{\frac{x-4}{x^2-9}}{1+\frac{1}{x-3}}$ by first finding the LCD of all “embedded” fractions.

Step 2: Subtract $\frac{1}{x-3}$ from your answer in Step 1 by first finding a common denominator.

(6.3(b) continued)

Follow the steps to simplify the expression below, then **refer to this worksheet while doing tonight's homework**.

$$\frac{\frac{1}{x+1}}{x - \frac{1}{x - \frac{1}{x}}}$$

Step 1: Simplify the complex fraction $\frac{1}{x - \frac{1}{x}}$ that is found in the denominator of the rational expression above.

Step 2: Plug your answer from Step 1 into the denominator of the expression and then simplify the resulting complex fraction by first finding the appropriate LCD.

Review of 6.1 – 6.4

1. Simplify $\frac{x^2 - x - 6}{x^2 + 3x - 18}$

2. Multiply $\frac{x^3 - 27}{4x^2 - 4x} \cdot \frac{4x}{x - 3}$

3. Add $5 + \frac{7}{x - 2}$

4. Simplify $\frac{x - \frac{4}{x + 6}}{\frac{1}{6 + x} + x}$

(6.1–6.4 review continued)

5. Divide $(28x^8y^3 - 14x^6y^2 + 3x^2y^2) \div (7x^2y)$

6. Subtract $\frac{2x-1}{x+6} - \frac{x+3}{x-2}$

7. Subtract $\frac{3}{x-2} - \frac{2}{x+2} - \frac{x}{x^2-4}$

(6.1–6.4 review continued)

8. Divide $\frac{3x^2 - 7x - 6}{3x^2 - 13x - 10} \div \frac{2x^2 - x - 1}{4x^2 - 18x - 10}$

9. Add $\frac{3}{7-x} + \frac{x-2}{x-7}$

10. Simplify $\frac{x+1}{x^2+x-2} - \frac{1}{x^2-3x+2} + \frac{2x}{x^2-4}$

6.6 Rational Equations

Steps for solving rational equations:

- (1) Make note of any x values that cannot be solutions because they would cause you to divide by 0.
- (2) *Clear the denominators* by multiplying both sides of the equation by the LCD of all fractions.
- (3) Solve the resulting equation.
- (4) Remove any impossible solutions (from Step 1).
- (5) Check the remaining solutions by plugging them into the original equation.

Solve the following rational equations.

1. $1 + \frac{4}{x-5} = \frac{2}{x^2-5x}$

2. $\frac{3}{x-5} + \frac{2}{x+5} = \frac{30}{x^2-25}$

3. $\frac{5}{8x} - \frac{7}{4} = \frac{3}{2x}$

(6.6 continued)

Remember, you can **solve** rational **equations**, but you can only **simplify** rational **expressions**.

4. Simplify $\frac{1}{x+2} + \frac{1}{x-2}$

5. Solve $5 = \frac{4}{x} + \frac{9}{x+1}$

6. Solve $\frac{2}{x+3} + \frac{x}{x-3} = \frac{-10}{x^2-9}$

7. Simplify $\frac{2}{x+3} + \frac{x}{x-3} + \frac{10}{x^2-9}$

3 Rational Formulas and Applications

- Monday:

- No class

- Tuesday:

- Topic: Rational Formulas (6.7(a))
- **Quiz:** 6.6
- Worksheets p. 22-23
- Homework: 6.7(a)

- Wednesday:

- Topic: Word Problems with Rational Formulas (6.7(b))
- **LAB:** Laser Beam Reflections
- Homework: 6.7(b)

- Thursday:

- Topic: Direct and Inverse Variation (6.8)
- **Quiz:** 6.7
- Worksheets p. 24-25
- **Worksheets p. 21-25 and LAB due at end of class**
- Homework: 6.8

6.7(a) Rational Formulas

Solve each formula for the specified variable.

1. Solve $\frac{V_1}{V_2} = \frac{P_1}{P_2}$ for V_1 .

2. Solve $A = \frac{1}{2}(b_1 + b_2)(h)$ for b_1 .

3. Solve $P = \frac{A}{1 + rt}$ for t .

4. Solve $F = \frac{9}{5}C + 32$ for C . (*This is the formula to convert Celsius to Fahrenheit.*)

5. Solve $V = \frac{1}{3}\pi r^2 h$ for h . (*This is the formula for the volume of a cylinder.*)